Weekly Report: Ultra-High-Pressure ODMR — Why should the anvil culet be [111]? Testing the $E = 0$ argument

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Question

A [111] culet puts one NV family exactly along the load axis, so that family keeps its C_{3v} symmetry and its transverse-strain splitting E vanishes; does that argument actually make [111] the better choice for ODMR sensitivity at 120 GPa, compared with a [100] culet where all four families are equivalent?

Hypothesis

If the axial family on a [111] culet has $E = 0$ and therefore no transverse-strain broadening, it should give the narrowest line and win despite carrying only 1/4 of the NV population, whereas [100] keeps all four families but pays a linewidth penalty $\propto |dE/dt|\,\delta t$ from the spread δt in deviatoric stress across the culet.

Evidence / Interpretation

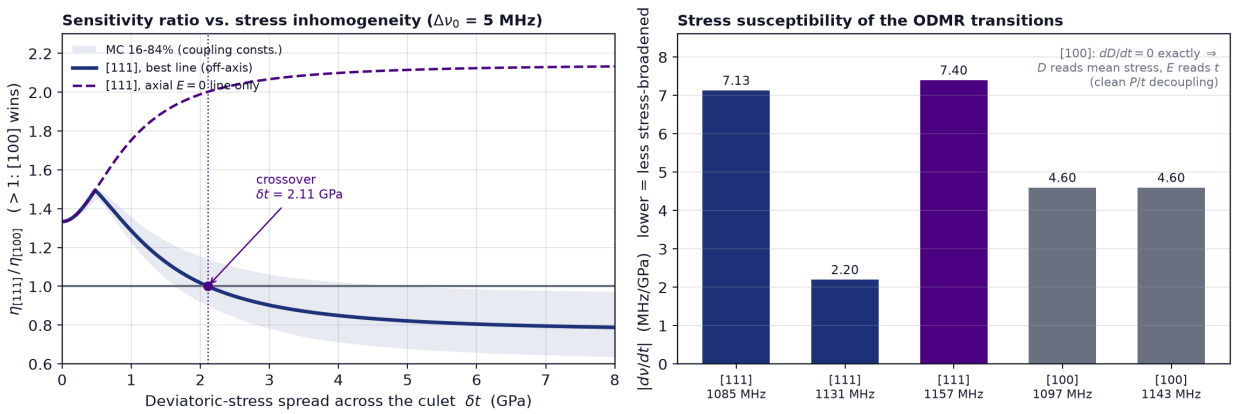

図 1: Left: $\eta_{[111]}/\eta_{[100]}$ vs. the deviatoric-stress spread δt, for the best [111] line and for the axial $E = 0$ line alone, with a Monte-Carlo band over the Barson spin-stress constants. Right: $|d\nu/dt|$ of every line; the $E = 0$ line is the steepest one.

- The symmetry argument itself holds exactly ($E = 0$ and $dE/dt = 0$ for the axial family), but that line is nevertheless the *most* deviatoric-stress-sensitive one in the problem, $|dD/dt| =$

$|2a_2| = 7.40\,\mathrm{MHz/GPa}$ versus $4.60\,\mathrm{MHz/GPa}$ for [100], and with only 0.375 of the PL dipping it loses to [100] by $1.33\times$ to $2.14\times$ at every δt.
- [111] does win, but above $\delta t \approx 0.42\,\Delta\nu_0$ (GPa per MHz of intrinsic width, i.e. 2.1 GPa for $\Delta\nu_0 = 5\,\mathrm{MHz}$) and through the *off-axis* families, whose line at $|d\nu/dt| = 2.20\,\mathrm{MHz/GPa}$ is the narrowest available — so the correct reason to pick [111] is the opposite of the usual one.

Gap / Failure

- The model scores only $\eta \propto \Delta\nu/(C\sqrt{R})$ at zero bias field, so it cannot see the genuine [111] advantage for vector magnetometry, where a field along the axial NV is unmixed by E.
- It also says nothing about the dominant real-world reason for [111] anvils — the higher compressive strength and the cleavage-plane geometry that decide whether the anvil survives 120 GPa at all.

Next Step

Fold the δt-dependent linewidths from this model into the background-aware sensitivity code so that the culet choice, the excitation wavelength and the background level are optimised together rather than one at a time.

References

1. M. S. J. Barson *et al.*, "Nanomechanical Sensing Using Spins in Diamond," Nano Lett. **17**, 1496 (2017) — spin-stress constants a_1, a_2, b, c.
2. P. Udvarhelyi *et al.*, Phys. Rev. B **98**, 075201 (2018) — spin-strain Hamiltonian conventions.
3. M. W. Doherty *et al.*, Phys. Rev. Lett. **112**, 047601 (2014) — NV under pressure.
4. Repository: `code/anvil_orientation.py`, `code/fig6_anvil_orientation.py` (this analysis).